Wildland surface fire **spread** modelling, 1990–2007. 1: Physical and quasi-physical models

AL Sullivan - International Journal of **Wildland Fire**, 2009 - CSIRO Publishing

… reason for the inclusion of his work in the current **review**. … to be used to **model wildland fire spread** across large spatial … of the fire itself; **mapping** of the **spread** of wildfires is haphazard …

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Applications of **simulation**-based burn probability modelling: a **review**

MA Parisien, DA Dawe, C Miller… - … of **wildland fire**, 2019 - CSIRO Publishing

… This method, termed burn probability (BP) modelling, takes advantage of fire **spread** … assess BP **model** accuracy. While it is tempting to compare a BP **map** with recent patterns of **wildfire** …

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DEVS-FIRE: Towards an integrated **simulation** environment for surface **wildfire spread** and containment

L Ntaimo, X Hu, Y Sun - **Simulation**, 2008 - journals.sagepub.com

… (HLA)compliant distributed M&S environment formed by **map**ping the DEVS-C++ system [14] to the … We **review** the basics of **wildfire behavior** [56] to set the ground for the DEVS-FIRE …

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A **review** of **wildland fire spread** modelling, 1990-present, 1: Physical and quasi-physical models

AL Sullivan - arXiv preprint arXiv:0706.3074, 2007 - arxiv.org

… Whereas FIRETEC was intended to be used to **model wildland fire spread** across large spatial scales, IUSTI … **Mapping** of the **spread** of wildland fires is haphazard and highly subjective. …

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Modelling smoke transport from wildland fires: a **review**

SL Goodrick, GL Achtemeier, NK Larkin… - … Journal of **wildland fire**, 2013 - connectsci.au

… Another grid **model** receiving attention for **wildland fire** related air quality issues is the … of the modelling studies cited in this **review** involves the **dynamics** of the buoyant phase of the …

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A systematic **review** of applications of machine learning techniques for **wildfire management** decision support

K Bot, JG Borges - Inventions, 2022 - mdpi.com

… decision-making, with a **forest fire susceptibility map** as an output. In [… **spread** is estimated using a deep convolutional inverse graphics network. In [125], the **prediction** of **wildfire spread** …

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Spatial **prediction** of **wildfire susceptibility** using field **survey** GPS data and machine learning approaches

O Ghorbanzadeh, K Valizadeh Kamran, T Blaschke… - Fire, 2019 - mdpi.com

… The regularization (C) parameter also influences the **behavior** of SVM. Thus, determining … For our future work, since such spatial **prediction** of **wildfire susceptibility maps** can illustrate …

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Progress and prospects for predicting **wildfire spread** through the wildland-urban interface

O Price, S Ondei, DMJS Bowman - International Journal of Disaster Risk …, 2025 - Elsevier

… Scientific understanding of **wildfire spread** into urban areas is … **review** the state of knowledge around modelling fire **spread** in … We suggest that our proposed **model** is feasible given …

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Wildland surface fire **spread** modelling, 1990–2007. 2: Empirical and quasi-empirical models

AL Sullivan - International Journal of **Wildland Fire**, 2009 - CSIRO Publishing

… present **review** as it is outside the scope of surface fire **spread**). … between the **model** and observations of rate of **spread** of a … The position of the flames in relation to the grid was **mapped** …

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An empirical-based **model** for predicting the forward **spread** rate of wildfires in eucalypt forests

MG Cruz, NP Cheney, JS Gould, WL McCaw… - … journal of **wildland fire**, 2022 - connectsci.au

… recent **wildfire** events. We aimed to evaluate the **model** against independent experimental fire and **wildfire** … of **Wildland Fire**, but were blinded from the peer-**review** process for this paper. …

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Representation and evaluation of **wildfire propagation** simulations

JB Filippi, V Mallet, B Nader - International Journal of **Wildland Fire**, 2014 - connectsci.au

… mathematical representation of a **wildfire simulation**, reviews the … , such as elevation, fuel distribution and wind **maps**, but … Far from being exhaustive, this **review** tries to present several …

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WRF-Fire: coupled weather–**wildland fire** modeling with the weather research and forecasting **model**

JL Coen, M Cameron, J Michalakes… - Journal of Applied …, 2013 - journals.ametsoc.org

… numerical weather **prediction model**. The fire module is a surface fire-**behavior model** that is two-way coupled with the atmospheric **model**. Near-surface winds from the atmospheric …

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Overview of **bushfire spread simulation** systems

P Johnston, G Milne, D Klemitz - **BUSHFIRE** CRC Project B, 2005 - naturalhazards.com.au

… **review** of the state of the art of **bushfire simulation** systems. We discuss the input data for a **bushfire** behaviour **model** … the ignition of the fire to obtain a **map** of fuel moistures based on the …

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A cellular automaton **model** of **wildfire propagation** and extinction

KC Clarke, JA Brass, PJ Riggan - … Engineering and Remote …, 1994 - research.fs.usda.gov

… -metre DigitaJ Elevation **Model** from the US Geological **Survey** (USGS)… **maps** can be produced. Far more intriguing as an application, however, is to use the **model** to predict the **behavior** …

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Forest fire spread simulating **model** using cellular automaton with extreme learning machine

Z Zheng, W Huang, S Li, Y Zeng - Ecological Modelling, 2017 - Elsevier

… **simulation** of **forest fire** spreading plays an essential role in designing quick risk **management** and … By integrating the ELM with the traditional **forest fire** CA framework, a new cellular …

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Wildfire risk modeling

S Oliveira, J Rocha, A Sá - Current Opinion in Environmental Science & …, 2021 - Elsevier

… **review** of the methods and approaches applied to **model wildfire** risk and its key components, in particular since 2018, and how they can contribute to **wildfire management**. …

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Modeling **wildfire spread** in wildland-industrial interfaces using dynamic Bayesian network

N Khakzad - Reliability Engineering & System Safety, 2019 - Elsevier

… **wildfire spread** in WIIs by combining dynamic Bayesian network and **wildfire behavior prediction**
… The developed methodology models the spatial and temporal **spread** of fire, based on …

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A real-time computer application for the **prediction** of fire **spread** across the Australian landscape

JR Coleman, AL Sullivan - **Simulation**, 1996 - journals.sagepub.com

… Models for the **prediction** of **wildfire spread** and **wildfire** behaviour across landscapes have
… Each perimeter point record includes the **map** coordinates of the point in Australian **Map** …

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An integrated approach for tactical monitoring and data-driven **spread** forecasting of wildfires

MM Valero, O Rios, C Mata, E Pastor, E Planas - Fire Safety Journal, 2017 - Elsevier

… and simulators able to **forecast** the fire **spread**. However, … , simulators' accuracy on fire **spread prediction** has proved insufficient. … automated **wildfire** monitoring with accurate fire **spread** …

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SPARK–A **bushfire spread prediction** tool

C Miller, J Hilton, A Sullivan, M Prakash - International Symposium on …, 2015 - Springer

… a probability **map** of fire **spread** over a given time period. The … **wildfire spread** models using a level set approach. Submitted to Environmental Modelling and Software (under **review**…

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